

On the Role of Sea Surface Temperature in the Clustering of Global Tropical Cyclone Formation

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ABSTRACT: This study examines the potential impacts of large-scale atmospheric circulations that are forced by sea surface temperatures (SST) on global tropical cyclone (TC) formation. Using the Geophysical Fluid Dynamics Laboratory (GFDL) global atmosphere and land surface model, version 4 (AM4), under different SST distributions, it is found that the east–west clustering of global TC formation is mainly governed by large-scale circulations in response to given SSTs, instead of direct ocean surface fluxes associated with zonal SST anomalies. Our zonally homogeneous SST simulations in the presence of realistic surface coverage show that TC clusters still emerge as a result of the breakdown of zonal circulations related to land–sea distribution, which produce specific “hotspots” for global TC formation. Sensitivity experiments with different climate warming scenarios and model physics confirm the persistence of these TC clusters in the absence of all zonal SST variations. These robust results offer new insights into the effects of large-scale circulation and terrain forcing on TC clusters beyond the traditional view of direct SST impacts, which are based on the direct alignment of the warmest SST regions and TC clusters. In addition, our experiments also capture internal variability of the global TC frequency, with an average fluctuation of 6–8 TCs at several dominant frequencies of ~3, 6, and 9 years, even in the absence of all SST interannual variability and ocean coupling. This finding reveals an intrinsic “noise” level of the global TC frequency that one has to take into account when examining the past and future trends in TC activity and their related significance or detectability.

SIGNIFICANCE STATEMENT: In this study, the clustering of global tropical cyclone (TC) formation is investigated, using global simulations under different idealized sea surface temperature (SST) distributions. Our results show that it is the response of the large-scale tropical circulations to SST anomalies that is mostly responsible for the clustering of global TC formation rather than surface flux differences. It is also found that the tropical atmosphere contains inherent fluctuations in the global TC frequency of 6–8 TCs every 3–9 years, even in the absence of all SST interannual and zonal variability. These results offer new insight into the role of tropical dynamics in governing TC climatology and suggest possible mechanisms underlying the clustering of global TC formation under different climate conditions.

KEYWORDS: Tropical variability; Climate variability; Tropical cyclones; Hurricanes/typhoons; Climate models

1. Introduction

Global tropical cyclone (TC) climatology has been well documented to possess distinct regions where TC formation is most favored. Within the band of latitudes where TCs typically form (between 20°S and 20°N), there are five major clusters of TC formation including the northwestern Pacific Ocean, eastern Pacific, North Atlantic Ocean, Indian Ocean, and southern Pacific basins. Among several environmental conditions favorable for TC formation such as vertical wind shear, large-scale circulation or tropical waves, sea surface temperature (SST) is often considered to be the key factor accounting for such clustering of global TC formation due to the strong dependence of surface fluxes on temperature (e.g., Gray 1968, 1982; McBride and Zehr 1981; Wang and Chan 2002; Chan 2005; Ramsay 2017). Indeed, direct comparison of the SST distribution and TC climatology shows good alignment between these two at the global scale. These specific SST anomaly patterns that accord with the TC

clusters across ocean basins are ultimately related to the land–sea distribution, which accounts for SST irregularity via oceanic circulations. Such a key role of the land surface arrangement in dictating SST climatology is also consistent with numerous aquaplanet simulations, which show that the clustering of global tropical cyclogenesis (TCG) is absent in all aquaplanet settings with zonally symmetric SSTs and/or solar insolation (e.g., Merlis et al. 2013; Rauscher et al. 2013; Chavas and Reed 2019; Walsh et al. 2020; Vu et al. 2021).

From the climatological perspective, the dominant role of SST in the clustering of global TCG is most manifested in the northwestern Pacific (WP) basin, which is often attributed to a large warm pool in this area and its extended season (e.g., Lander and Guard 1998; Cheung 2004; Vecchi and Soden 2007; Ramsay 2017; Sobel et al. 2021). While this WP warm pool varies from year to year due to El Niño–Southern Oscillation (ENSO), the WP is generally warmer than all other basins and explains the dominance of the WP TC cluster when coupled with active intertropical convergence zone (ITCZ) or monsoon activities (e.g., Wang and Chan 2002; Emanuel and Nolan 2004; Cheung 2004; Chan 2005; Camargo and Sobel 2005;

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Camargo et al. 2007; Vecchi and Soden 2007; De Deckker 2016).

In the eastern Pacific, the highest density of TCG per surface area is, however, more related to the abundance of terrain-induced tropical disturbances (Zehnder et al. 1999; Molinari et al. 1997; Wang and Magnusdottir 2006; Halverson et al. 2007; Kieu and Zhang 2008, 2009). In contrast, the disturbance-generating mechanism in the North Atlantic (NA) basin is typically associated with the African jet, which has been shown to be the source of easterly waves that host TCG (e.g., Avila and Pasch 1992; Molinari et al. 2000; Schreck et al. 2012; Dunkerton et al. 2009; Wang et al. 2019). While these different TCG pathways all operate with different efficiency in different basins, observational and modeling studies have confirmed that the ITCZ is overall the most efficient in creating tropical disturbances for TC formation, especially in the WP basin (e.g., Gray 1982; Ritchie and Holland 1999; Molinari et al. 1997; Dickinson and Molinari 2002; Camargo and Sobel 2005). These tropical disturbances, when superimposed on a warm SST area, produce a higher chance of TC development and explain the existence of different TCG clusters.

The coupling between the ITCZ and SST anomalies that modulates TC climatology highlights the special role of air–sea interaction in global TC formation. Unlike the development of a single storm, for which SST plays a vital role via surface flux and energy conversion efficiency (Emanuel 1986), SST plays dual roles at the global scale. On the one hand, SST directly controls surface fluxes via the wind-induced surface heat exchange mechanism that governs the thermodynamics and dynamics of individual storm feedback (herein referred to as the *direct* SST effect). On the other hand, SST plays a broader role through its impacts on large-scale atmospheric thermodynamics and circulations, which can affect environmental conditions for TC development (herein referred to as the *indirect* SST effect). Various modeling studies have shown that the seasonal migration of the ITCZ and SST patterns are strongly related, supporting the broader indirect role of SST in tropical dynamics (e.g., Neelin and Held 1987; Lindzen and Nigam 1987; Merlis et al. 2013; Shaevitz et al. 2014; Ramsay 2017; Vu et al. 2021; Geen et al. 2020; Vidale et al. 2021). For example, SST can produce large-scale pressure gradients that contribute to low-level convergence in the tropics and govern the tropical circulations as shown in Lindzen and Nigam (1987). From this perspective, a natural question of what are the relative roles of the direct SST effect and the indirect SST effect in the clustering of global TC formation is still elusive.

The main objective of this study is to examine to what extent SST can govern the clustering of global TC formation. By carrying out global simulations with zonally homogeneous SST in the presence of realistic land surface condition, we can explore the direct and indirect roles of zonal SST distribution in the TCG climatology. In addition, we will examine the large-scale atmospheric factors that are responsible for the clustering of global TC formation under different SST settings.

With that, the rest of this work is organized as follows. Section 2 provides details of model description and experiment designs.

Section 3 discusses main results, and sensitivity analyses are provided in section 4. Main concluding remarks are then given in the final section.

2. Method

a. Model description

To capture the large-scale atmospheric dynamics without lateral boundary condition issues as in area-limited models, the Geophysical Fluid Dynamics Laboratory (GFDL) global atmosphere and land surface model, version 4 (AM4; Zhao et al. 2018a,b), is used in this study. AM4 has been used in GFDL's latest physical climate model (CM4; Held et al. 2019), Earth system model (ESM4; Dunne et al. 2020), and seasonal-to-decadal prediction system [Seamless System for Prediction and Earth System Research (SPEAR); Delworth et al. 2020]. It uses a hydrostatic version of the FV3 finite-volume cubed-sphere dynamical core, which employs a cubed-sphere topology with 96×96 grid boxes per cube face and corresponds to roughly 100-km horizontal resolution.

The default AM4 physical packages include the GFDL radiative transfer (Freidenreich and Ramaswamy 2011), topographic gravity wave drag (Garner 2005), a double-plume model representing shallow and deep convection (Zhao et al. 2018a), aerosol–cloud interaction (Ming et al. 2007), single-moment cloud microphysics (Rotsteyn 1997), and boundary layer parameterization (Lock et al. 2000) with the neutral drag formulation over oceans based on the Coupled Ocean–Atmosphere Response Experiment, version 3.5 (COARE3.5), consortium (Edson et al. 2013). AM4 uses the GFDL land model (LM4.0), which is based on the GFDL LM3.1 described by Milly et al. (2014). Calibration and verification of AM4 for a range of climate simulations within the framework of phase 5 of the Coupled Model Intercomparison Project (CMIP5) and CMIP6 have been reported by Zhao et al. (2018a,b), which demonstrate its capability in reproducing reasonable large-scale climatologies such as mean circulations, storm-track transition, and tropical dynamics as expected.

Specifically for our global TC simulation in this study, Zhao et al. (2018a,b) demonstrated that AM4 is capable of reproducing well the global TCG distribution, seasonal variability along with the total annual TC number. While AM4 has some low bias in the distribution of TCs in the North Atlantic basin, it performs as well as or better than many global climate models under similar resolutions (Shaevitz et al. 2014; Zhao et al. 2018a), with an overall highest correlation between the model-simulated and observed TC distributions in the WP basin. Detailed evaluations of AM4 for historical simulations can be found in Zhao et al. (2018a,b).

b. Experiment design

To focus on our main objective of examining the direct and indirect role of SST in the clustering of global TCG, the standard climate settings of the AM4 similar to Zhao et al. (2018a,b) were chosen. A control climate experiment (CTL) was then run for 30 years. The model is forced by monthly varying climatological SSTs and sea ice concentrations averaged over the 1981–2014

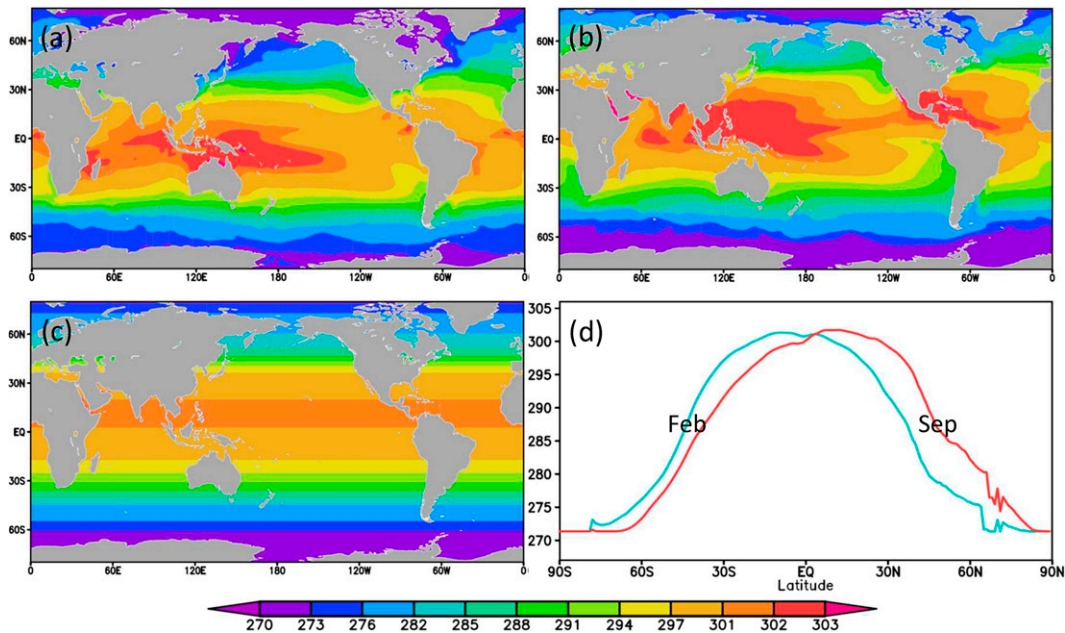


FIG. 1. (a) The SST distribution (shaded; K) averaged from 1981 to 2014 that is used in the CTL simulation for (a) February and (b) September; (c) zonally averaged SST used in the ZHT simulation using the SST data as in (b); (d) the zonal mean SST profiles in the ZHT simulation for February (blue) and September (red).

period with radiative gases and aerosol emissions fixed at the year 2010 condition. Our design of 30-yr simulation is such that the first year is treated as a spinup and the remaining years of the simulation are used to construct TC climatology and related statistics. Such a long 30-yr simulation is necessary to ensure that the model can develop consistent climatology with a given SST forcing and model physics.

Note that, unlike historical simulations that include the interannual variability of SST climatology, our control simulation has *seasonal but no interannual variability of SST during the entire course of the simulation*. With the 30-yr simulation, the TC statistics are therefore sufficiently stable and in accordance with such prescribed seasonal SST without the effects of interannual SST variability. Because this SST distribution is obtained from a climatological average, its overall spatial variation is well preserved, thus retaining the main features of SST climatology for our purposes as shown in Figs. 1a and 1b.

Starting from the CTL simulation, a new experiment with a different seasonal SST distribution was carried out. For this experiment, the SST distribution in the CTL simulation was first zonally averaged and then assigned to every ocean point at each latitude. By applying this process over the entire domain, a new SST distribution was obtained, which is now zonally homogeneous but still varies with latitude and season (herein referred to as ZHT; see Figs. 1c,d). This approach removes the zonal variation of SST and allows one to isolate the influence of SST zonal variation on the clustering of global TCG beyond the common aquaplanet framework that was widely used in previous studies (e.g., Rauscher et al. 2013; Merlis et al. 2013; Chavas and Reed 2019; Vu et al. 2021;

Walsh et al. 2020) or idealized simulations with realistic land surface conditions (Geen et al. 2018; Chiang et al. 2020).

Along with the ZHT experiment, additional sensitivity experiments were conducted to examine the robustness of the results obtained from the AM4 model. In the first group of sensitivity experiments, different global warming scenarios were applied to the ZHT configuration including (i) globally homogeneous warming at all model grid points by 2 K, (ii) polar warming for which the polar regions are warmed up by 2 K and the warming magnitude is gradually reduced to zero at the equator, and (iii) tropical warming for which the tropical region is warmed up by 2 K and the warming gradually decreases to zero at the polar regions (Fig. 2). These experiments

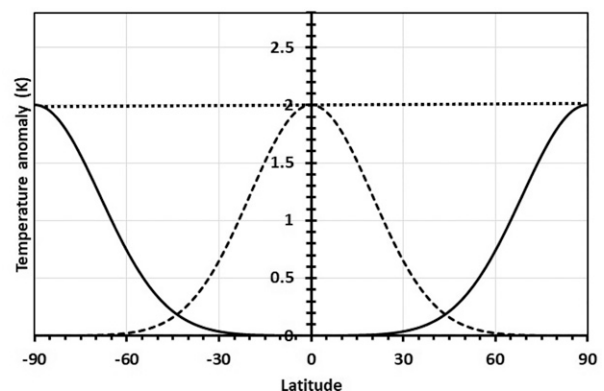


FIG. 2. Profiles of three different global warming anomalies for sensitivity experiments including globally homogeneous warming (dotted), polar warming (solid), and tropical warming (dashed).

TABLE 1. Descriptions of the AM4 experiments.

Expt	Remarks
CTL	A 30-yr control simulation using the seasonal SST climatology averaged from 1980 to 2010 as in Zhao et al. (2018a)
ZHT	Similar modeling settings as for the CTL but using the zonal average of the SST in the CTL experiment for the ocean area; this zonally averaged SST maintains the seasonal variability but not interannual variability
Tropical warming	Similar to ZHT but adding a maximum warming of 2 K to the tropical region and the warming is gradually reduced with higher latitudes, approaching zero at the polar regions
Polar warming	Similar to ZHT but adding a maximum warming of 2 K to the polar regions and the warming is gradually reduced with lower latitudes, approaching zero at the equator
Constant warming	Similar to ZHT but adding a constant warming of 2 K to all model grid points
DCTL	Similar to CTL but with the reduced convective detrainment (i.e., stronger deep convection) parameter
DZHT	Similar to ZHT but with the reduced convective detrainment (i.e., stronger deep convection) parameter

are aimed at exploring how different warming patterns could affect the global clustering of TC formation.

In the second group of sensitivity experiments, we reduce the lateral mixing rate in AM4's deep cumulus parameterization scheme by $\sim 20\%$ such that the influence of deep convection on large-scale tropical dynamics can be investigated. This reduction of lateral mixing generally enhances deep convection in both the CTL and ZHT experiments, which is known to be critical in modeling not only large-scale circulation but also storm-scale development (e.g., [Hart et al. 1990](#); [Song et al. 2008](#); [Zhao et al. 2009](#)). Thus, it might play a key role in the clustering of global TC formation. Table 1 summarizes the configurations for all simulations in this study.

Using the model output from these AM4 simulations, a tracking algorithm similar to that used in [Zhao et al. \(2009, 2018b\)](#) was employed to detect TC formation. This tracking algorithm is based on several standard properties of TCs including 850-hPa relative vorticity ($>1.6 \times 10^{-4} \text{ s}^{-1}$), upper-level warm core anomaly ($>1^\circ\text{C}$), the maximum surface wind ($>17 \text{ m s}^{-1}$), and the minimum central surface pressure, which are applied between 25°S and 25°N . In addition, a model vortex will be categorized as a hurricane if its maximum surface wind at any point during its entire trajectory exceeds 32 m s^{-1} . While different tracking algorithms could give different total TC counts as pointed out in previous studies (e.g., [Horn et al. 2014](#); [Zarzycki and Ullrich 2017](#); [Bourdin et al. 2022](#)), our focus here is on the relative difference in global TCG, seasonal variability, as well as change in the large-scale environment between the CTL and ZHT experiments. So long as the same tracking criteria are applied to all experiments, the relative difference in TC clusters among these experiments is meaningful and comparable.

A few remarks about our zonally homogeneous SST configuration are given here. First, SST in our ZHT experiment is treated as an external forcing rather than dynamically coupled to the ocean beneath. While this uncoupled setting is favored in our experiments such that the zonally homogeneous SST can be maintained, some large-scale atmospheric modes related to ocean dynamics such as ENSO or Pacific decadal oscillation cannot develop in the CTL and ZHT experiments. The lack of coupled ocean–atmosphere climate modes certainly

renders the TC climatology and its variability in this study incomplete, yet it is this same lack of interannual climate modes that allows us to also more easily isolate the response of the atmosphere to different SST forcings.

Second, the SST field chosen in this study was taken from the 30-yr (1980–2010) average, which may not capture some enhanced favorable conditions for TC development from year to year due for example to different ENSO phases. We note, however, that the main objective of this study is to compare the TCG distribution between a realistic SST climatology and zonally homogeneous SST under the same land surface coverage. As long as the AM4 model can capture the first-order effect of the prescribed SST forcing, such a relative comparison among different TC climatologies is reasonable and will be assumed in all analyses hereinafter.

3. Results

a. Model TC climatology

To have a broad picture of TC climatology in the AM4 model under the controlled SST forcing, Fig. 3 compares global TC frequency that is obtained from the CTL and ZHT simulations. One notices first that the global TC frequency varies from year to year in both experiments, with no clear trend in the absence of SST interannual variability. Also, the global TC count in the ZHT experiment is about 20% more than what obtained in the CTL experiment, even when the ZHT maximum SST is slightly smaller than that in the CTL due to the zonal average. Such an increase in the total TC count in the ZHT experiment is to some extent expected, because of the larger area of warm SST over the entire tropical region.

Further analysis of the TC seasonality shows that the increase in global TC frequency in the ZHT experiment is mostly due to more TC formation over the Northern Hemisphere (NH) summer season as compared with that in the CTL experiment. In the Southern Hemisphere (SH; Figs. 3c,d), ZHT and CTL show little difference in both TC count and seasonality. This substantial difference in TC count between the two experiments is related to the more organized ITCZ over the Pacific Ocean as will be shown later, along with a warmer NA basin in the ZHT relative to the CTL experiment.

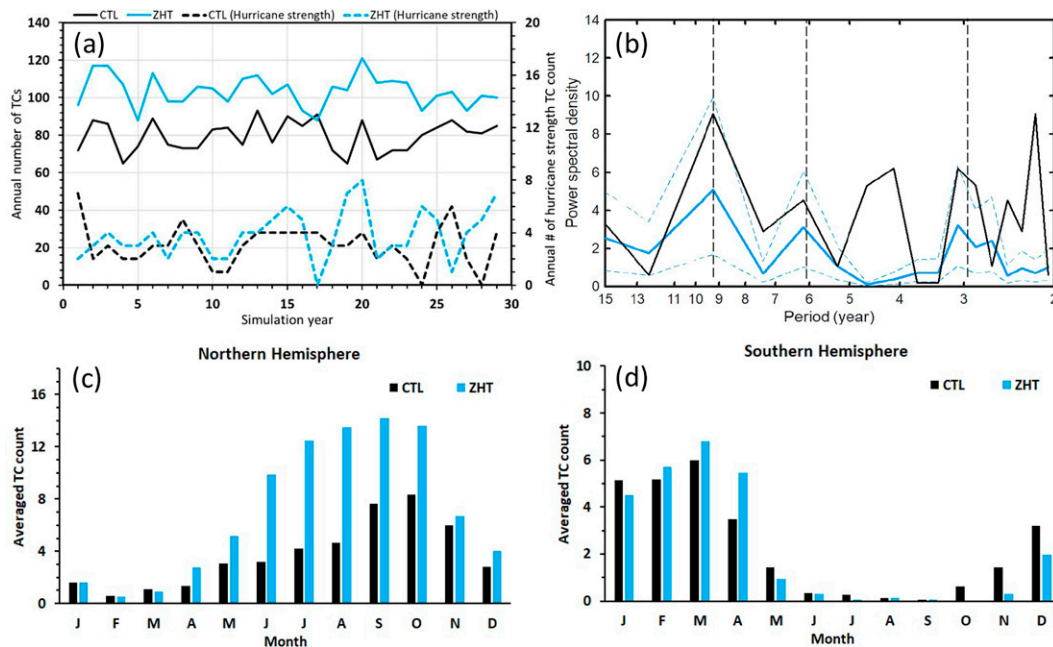


FIG. 3. (a) Time series of the global TC counts as obtained from the CTL (black) and ZHT (blue) experiments during the 30-yr simulation periods; (b) the power spectral density of the global TC count time series in (a). Seasonal distribution of annually averaged TC counts from the CTL (black) and ZHT (blue) simulations for the (c) NH and (d) SH. The blue dashed lines in (b) represent the upper and lower confidence bounds for ZHT at 90% confidence interval using all sensitivity experiments shown in Fig. 10a. Note that the power spectral density obtained from discrete-time Fourier transform for the TC time series shown in (b) has no unit.

It is of interest that while the total number of TCs is significantly different between the ZHT and CTL experiments, the number of hurricane-strength TCs is still about the same between the two experiments (Fig. 3a). The smaller value of the warmest SST in the ZHT experiment is evidently offset by a larger area of warm SST as well as more organized ITCZ structure under zonally uniform SST condition (cf. Figs. 5b,d, described in more detail later). Thus, the number of strong TCs in the model is determined not only by the SST magnitude but also by how conducive the large-scale conditions are to support TC development all the way to the maximum intensity limit. In this regard, more TC formation does not necessarily imply a higher number of strong TCs as seen in Fig. 3a.

Also of importance from Fig. 3 is the intrinsic variation from year to year in the global TC frequency, which is apparent in both the CTL and ZHT experiments even in the absence of all interannual SST variability and ocean coupling. Specifically, an average fluctuation of 6–8 TCs at several common frequencies of 3, 6, and 9 years for the given SST distribution obtained in both the CTL and ZHT experiments appears to be a manifestation of the tropical atmosphere's intrinsic variability. From a large-scale perspective, this internal variation of global TCG should be related to the characteristics of the atmosphere under the terrain-induced forcing in both the CTL and ZHT experiments. While a physical explanation for the nature of this intrinsic variability of the tropical atmosphere as well as its statistical significance is beyond the

scope of our analyses, the frequencies and amplitude of these internal fluctuations are consistent among all sensitivity experiments (cf. Figs. 10a and 12a, described in more detail later). Thus, these internal fluctuations may indicate an intrinsic noise level of the annual TC numbers that one needs to take into account when detecting the signal of TC frequency change in the past historical data or future climate.

In the real climate system where SST variability and other climate-related modes related to ocean–atmosphere coupling or other surface forcings are present, the variability of TC frequency is much more complicated, because the coupled ocean–atmosphere variability may drive some further variability in large-scale environments required for TCG. However, the CTL and ZHT experiments in this study could provide at least an indication of how much TCG variability can arise strictly from intrinsic atmospheric variability under realistic surface conditions that aquaplanet simulations could not capture.

b. Global TC formation clusters

For the spatial distribution of global TCG, Fig. 4 shows the climatology of TCG obtained from the CTL and ZHT experiments. Here, the TCG climatology is defined as a collection of TC centers that are *first* detected from our tracking algorithm during the 30-yr simulations. By dividing the model domain into boxes of $5^\circ \times 5^\circ$ and counting all TC centers forming within each box, the distribution of TCG can be then constructed from the model output over the entire domain.

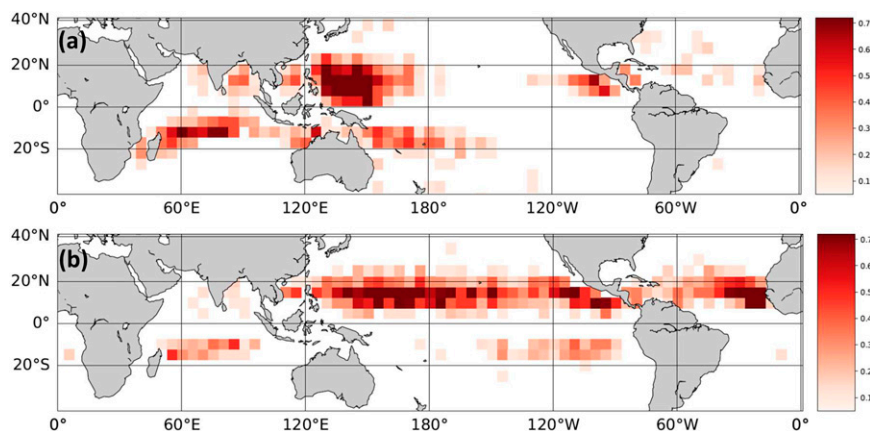


FIG. 4. Distribution of TC formation density (shaded; number yr^{-1}) as obtained from the 30-yr simulation of the (a) CTL and (b) ZHT experiments. Here, the TC formation density unit is the annual number of TCs forming per unit area, where the unit area is defined as a box of $5^\circ \times 5^\circ$.

Overall, AM4 reproduces the main clusters of TC formation in the CTL experiment, even at the coarse resolution of ~ 100 km as shown in Fig. 4a (see also Fig. 24 of Zhao et al. 2018a). Specifically, we recover what Zhao et al. (2018a) simulated, with a reasonable distribution of TC formation in the WP, southern Pacific, Indian Ocean, and eastern Pacific as well as the TCG seasonality (e.g., Zhao et al. 2018a, Fig. 25). Note, however, that the AM4's TCG climatology appears to be significantly underestimated in the NA basin (see the observed distribution, e.g., in Fig. 1 in Bloemendaal et al. 2020). Such difficulty of AM4 in reproducing TCG in the NA basin turns out to be common in many global climate models (e.g., Zhao et al. 2009; Shaevitz et al. 2014; Vidale et al. 2021). Among current global models, the GFDL High Resolution Atmospheric Model (HiRAM) tends to be the best one in the NA basin, yet this model is more prone to TCG in the entire tropics as compared with other global models at the same resolution (e.g., Harris et al. 2016). Of course, our CTL experiment has no SST interannual variability, and so it is not expected that its TCG distribution fully matches the real TCG distribution. However, the fact that AM4 and other global models have a similar underestimation of TCG in the NA basin suggests some systematic issues with the current models in capturing TC development in this specific basin, which are beyond the scope of this study.

Relative to the CTL experiment, Fig. 4b shows the density of TCG obtained from the ZHT experiment, which captures several noticeable features. First, the dominant TC cluster in the WP basin no longer exists in the ZHT experiment after the WP warm pool is removed. Instead, global TC formation now forms a more distinct band as expected for a zonally homogeneous SST distribution, especially in the northern Pacific Ocean and North Atlantic. In the SH, a somewhat similar TCG band pattern is also observed, albeit with significantly lower TCG density as compared with the NH.

Second, the existence of TCG clusters is still apparent in the ZHT experiment despite the use of zonally homogeneous

SST, even after taking a 30-yr average. For example, the TCG band in the northern Pacific has a nonuniform arrangement of TC formation along the band, with a new cluster near the central Pacific, followed by one in the eastern Pacific, and another one in the North Atlantic basin. While the northeastern Pacific and Indian Ocean clusters are somewhat similar in both the CTL and ZHT experiments, the northwestern, north-central, and southern Pacific show much more changes in the cluster locations. This is a noteworthy result because removing the zonal SST anomalies does not eliminate TCG clusters in the open ocean as in the aquaplanet simulations. Of course, the disappearance of the TCG cluster in the WP basin after the WP warm pool is removed indicates that this warm SST anomaly is the key factor for the clustering of TCG in the WP. However, the formation of new TCG clusters across ocean basins in the ZHT experiment strongly suggests some large-scale controls to the global TCG clustering in the presence of realistic land surface, rather than the direct effects of zonal SST anomalies.

Last, the asymmetry in both TCG density (Fig. 4) and frequency distribution (Fig. 3) between the SH and NH is significantly stronger in the ZHT experiment. This asymmetry is intriguing, given almost the same SST magnitude in both hemispheres (Fig. 1d). The fact that this profound asymmetry in TCG between two hemispheres is absent in aquaplanet experiments (e.g., Chavas and Reed 2019; Vu et al. 2021; Walsh et al. 2020) suggests the potential effects of the land surface on not only large-scale atmospheric dynamics but also the meridional SST profiles. In fact, our sensitivity experiments with different meridional SST profiles between the SH and NH confirms that the SST meridional profile is key factor in causing the ITCZ to preferably stay in the NH during the AM4 simulation (not shown), consistent with the studies by, for example, Donohoe et al. (2013) or Byrne et al. (2018).

To help to understand such larger asymmetry in the TCG climatology in the ZHT experiment, Fig. 5 compares the patterns of the ITCZ and SST during the summer and winter

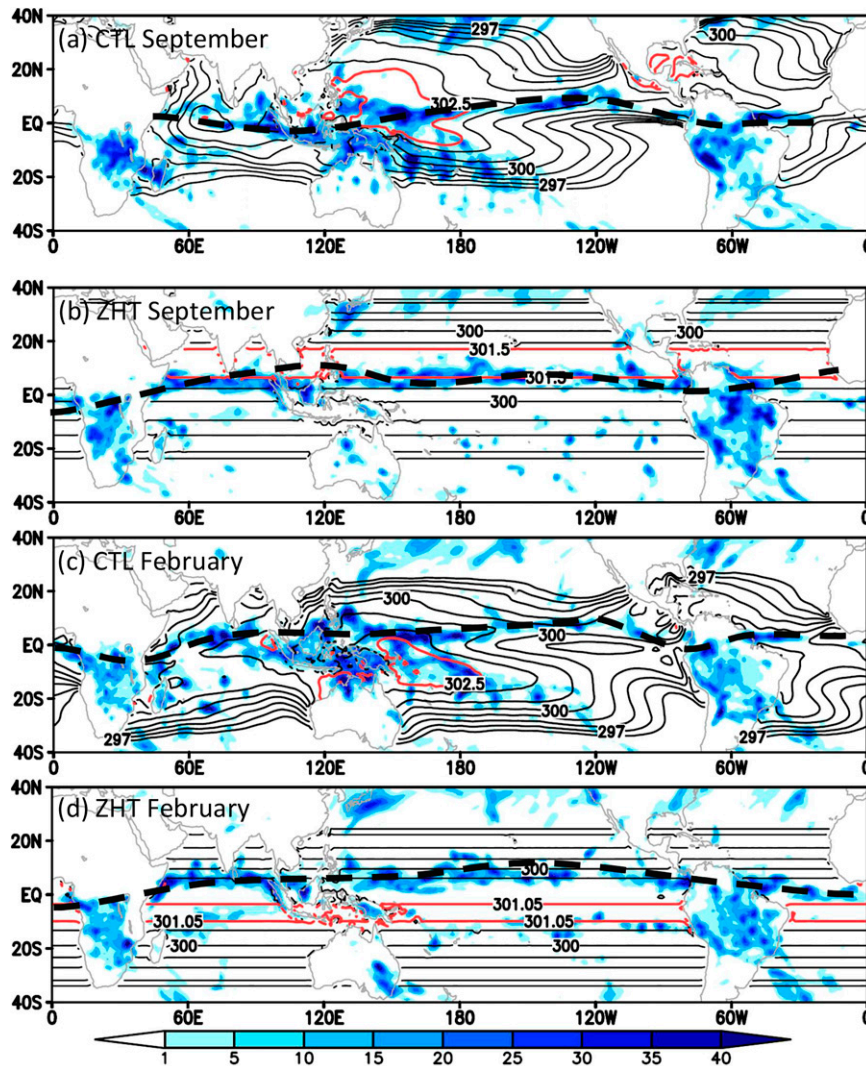


FIG. 5. Global distribution of precipitation (shaded; mm day⁻¹) and SST (contours at interval of 1 K) as obtained from (a) CTL in September, (b) ZHT in September, (c) CTL in February, and (d) ZHT in February. Thick dashed lines denote the ITCZ location, and red solid contours denote the warmest SST areas for each season and experiment.

seasons for the CTL and ZHT experiments. Here, we define the ITCZ as a zonally oriented band of maximum precipitation in the tropical region ($\pm 10^\circ$) as in [Geen et al. \(2020\)](#). During the main NH summer season, the alignment of the warmest SST region and the ITCZ produces a particularly favorable environment for TC formation ([Fig. 5a](#)), thus accounting for the dominance of TCG in both the CTL and ZHT simulations. Furthermore, the main TCG cluster in the WP basin is apparent in the CTL experiment due to the SST warm pool here as expected. For the ZHT experiment, the WP warm pool no longer exists and the alignment of the ITCZ and the warmest SST band over the entire Pacific Ocean may therefore explain the extended TCG band in the NH summer ([Figs. 4b](#) and [5b](#)). The disappearance of the WP warm pool in the ZHT experiment also results in weakened

monsoon activities in both the Indian Ocean and WP basin (cf. [Figs. 7g,h](#), described below) and leads to reduced TC formation in these areas as seen in [Fig. 4b](#).

During the main SH summer (February), one notices, however, that the warmest SST band in the ZHT experiment migrates to 5° – 10° S, yet the ITCZ is still located in the NH around 4° – 5° N ([Fig. 5d](#)). This disconnection between the ITCZ and the warmest SST band in the ZHT experiment may help explain why TCG in the SH summer is much less effective than that in the NH summer. That the ITCZ could still produce TCG during the NH winter in the ZHT experiment is likely caused by the ITCZ breakdown process (e.g., [Ferreira and Schubert 1997](#); [Wang et al. 2010](#)), yet the effectiveness of this breakdown mechanism as compared with the warmest SST effects is unclear. Note that such a disconnection between the

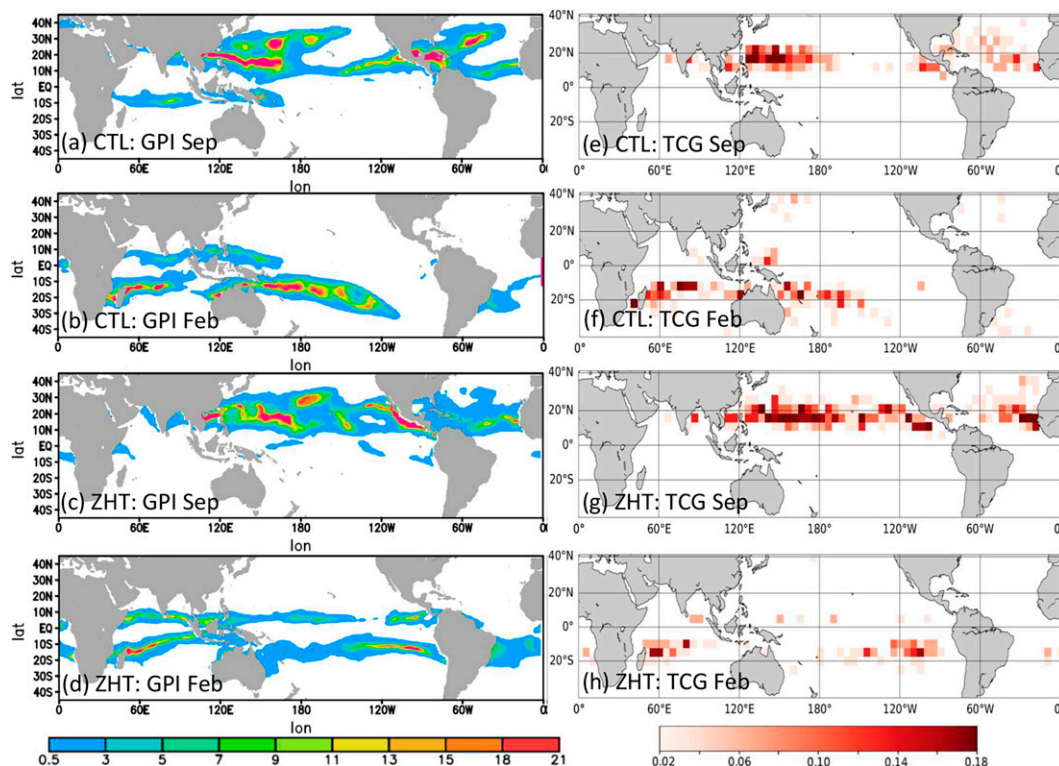


FIG. 6. Averaged distribution of the (a)–(d) GPI (shaded) and (e)–(h) the corresponding TCG density as obtained from (top) CTL September, (top middle) CTL February, (bottom middle) ZHT September, and (bottom) ZHT February.

ITCZ and the warmest SST is less critical in the CTL experiment, because the WP warm pool still covers part of the ITCZ and allows for a strong ascending branch of the Walker circulation in the southwestern Pacific Ocean (Fig. 5c), thus promoting more TC formation in the SH during January–March. In the ZHT experiment, the disconnection between the ITCZ and the warm SST band is critical, because it essentially removes almost entirely the WP cluster.

Given such different roles of the ITCZ and SST in the TC distribution, it is natural to quantify further how the favorable environmental conditions for TCG change between the CTL and ZHT experiments. For this, we use the genesis potential index (GPI; Emanuel and Nolan 2004) for additional analyses of the large-scale environments for TCG. Unlike the general use of GPI to model the basinwide averaged characteristics of TC climatology (Emanuel and Nolan 2004; Bruyère et al. 2012; Camargo et al. 2009), our use of GPI herein is to examine the spatial distribution of favorable conditions for genesis. This is a reasonable approach, because our 30-yr simulations are sufficiently long to ensure consistency with a given SST distribution. As described in section 2, the imposed SST contains seasonal variability but no interannual variability, thus producing useful statistics of TCG spatial variability for our purpose.

Overall, Fig. 6 shows that GPI could capture reasonably well the global picture of TC clusters in the CTL experiment. Indeed, a comparison of TCG density and GPI in the CTL

experiment (Figs. 6a,b,e,f) shows a good alignment of TC clusters and high GPI areas in all ocean basins, except for the North Atlantic, South Pacific, and the Gulf of Mexico where TCG density does not seem to correspond well to the large GPI. This mismatch between GPI and TCG density indicates that there are some other additional factors contributing to TCG that GPI is alone not able to fully capture. Unlike the CTL experiment, the consistency between GPI and TCG density is more apparent in the ZHT experiment, including the NA basin (Figs. 6c,d,g,h). The ZHT experiment actually captures also a secondary GPI band during February (Fig. 6d) associated with the ITCZ in the NH, which allows for additional TC formation in the NH even during the boreal winter season. This additional TC formation in the NH during the boreal winter contributes to the increased TCG asymmetry between two hemispheres as seen in Figs. 3c and 3d.

Of course, GPI is an empirical combination of several different large-scale factors that attempt to describe the overall environment needed for TCG. To specifically explore which GPI factor contributes the most to the shift of TCG clusters between the CTL and ZHT experiments, Fig. 7 shows further all four individual GPI factors including vertical wind shear (VWS), the maximum potential intensity (MPI), 700-hPa relative humidity (RH), and 850-hPa absolute vorticity for the Pacific Ocean. Here, we focus on the Pacific Ocean region because the shift of TCG clusters is most apparent here in both hemispheres as shown in Figs. 4 and 6, thus providing a

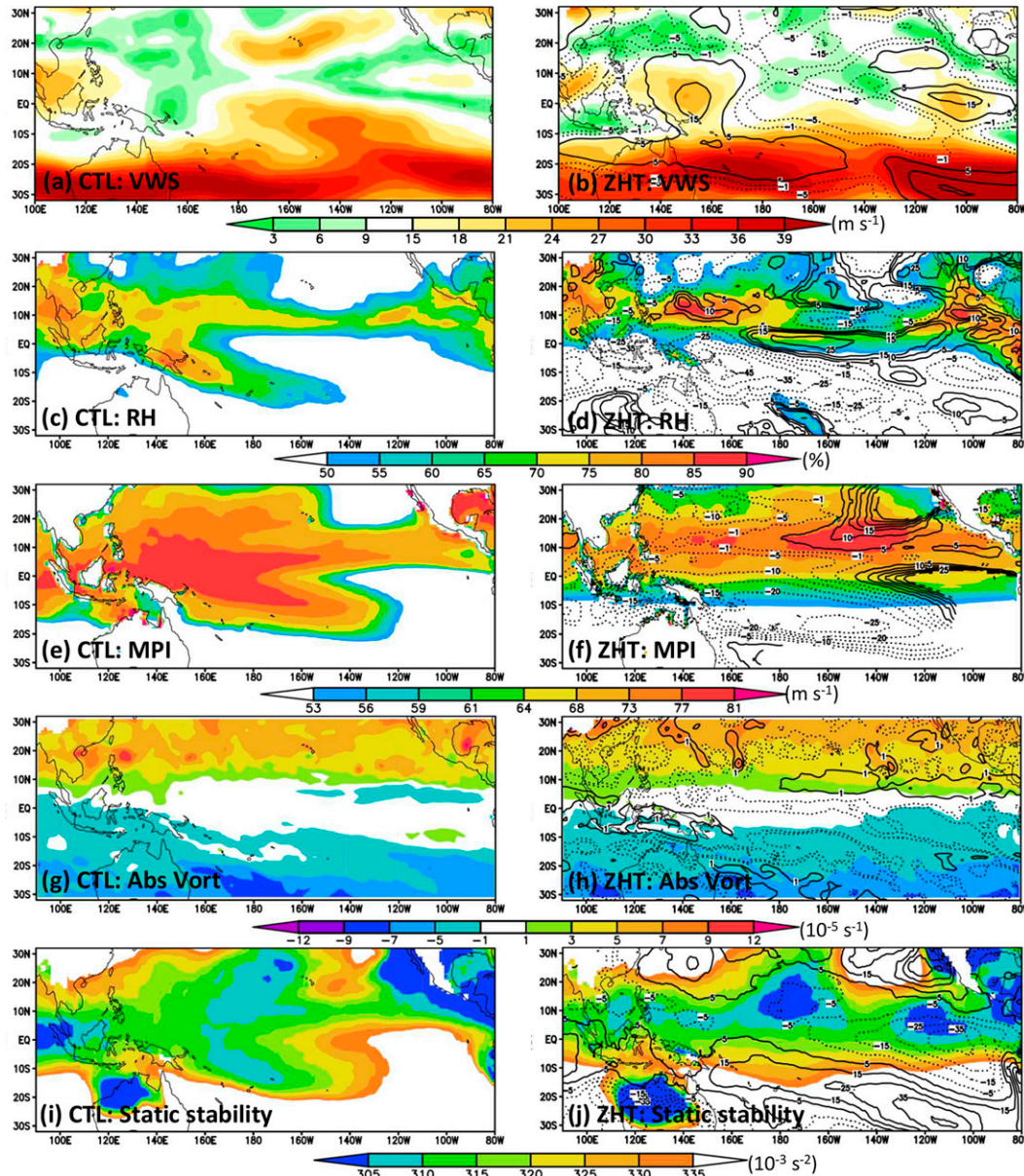


FIG. 7. Horizontal distribution of (a),(b) 850–200 hPa vertical wind shear (shaded; m s^{-1}); (c),(d) relative humidity (shaded; %); (e),(f) maximum potential intensity (shaded; m s^{-1}); (g),(h) the absolute vorticity (shaded; 10^{-5} s^{-1}); and (i),(j) tropospheric static stability (shaded; 10^{-3} s^{-2}) during September for the (left) CTL and (right) ZHT. Black solid and dotted contours in the right-hand panels respectively denote the positive and negative difference $\text{ZHT} - \text{CTL}$ for each corresponding GPI factor.

better way to examine the changes in the large-scale environment that cause the shift in TCG.

For the CTL experiment, all factors appear to play their role in the clustering of TCG in the Pacific Ocean. This is especially clear in the WP basin, where all factors display very favorable conditions associated with the warmest SSTs, including high potential intensities, a narrow zone of vorticity-rich environment related to monsoon activity, a moist lower troposphere due to strong low-level convergence, and low VWS. These favorable conditions explain why GPI is highest

in the WP basin in the CTL experiment, which is also consistent with the observed high TCG density in this basin during the boreal summer (Figs. 4 and 6).

Looking at the ZHT experiment, a somewhat different picture for these large-scale GPI factors is obtained. First, a new low-shear area along with higher RH now develops near the central Pacific as seen in Figs. 7b–d, which leads to the emergence of high GPI and a new TCG cluster near the central Pacific in the ZHT experiment. Note that ZHT captures, however, a reduction in both MPI and the 850-hPa absolute

vorticity in the same area near the central Pacific. The fact that a new TCG cluster could still form near the central Pacific thus suggests that MPI and the 850-hPa absolute vorticity play a less significant role in the clustering of TCG as compared with VWS and RH in the ZHT experiment.

Second, one may expect to see a horizontal band for MPI across the Pacific Ocean in the absence of SST zonal variation, yet ZHT displays an area with the largest MPI located in the east-central Pacific. Such a zonal variation of MPI in the ZHT experiment is probably due to changes in the atmospheric structure (Figs. 7i,j) and related convective available potential energy (CAPE), which leads to another region with high GPI in the eastern part of the central Pacific (cf. Fig. 6c). It should be mentioned that MPI depends not only on SST but also explicitly on CAPE and atmospheric stability (Kieu and Zhang 2018). Thus, the emergence of the elevated MPI areas in a zonally uniform SST setting (Fig. 6f) must be linked to the changes in the atmospheric large-scale thermodynamics and circulations beyond the direct SST influence.

It is also apparent in Fig. 7 that the high RH area and the vorticity-rich environment related to monsoon trough in the WP basin no longer exist in the ZHT experiment, and so the entire high GPI band in the eastern Philippine Sea no longer exists (cf. Figs. 6a,c) after removing of the WP warm pool. For example, Fig. 7h shows that the entire high-vorticity band from the Bay of Bengal to the eastern Philippine Sea, where the monsoon is climatologically most active (e.g., Wang and LinHo 2002; Beattie and Elsberry 2012), is removed in the ZHT experiment. Similarly, the decrease in RH in the WP basin also contributes to the disappearance of the WP cluster here, thus highlighting the different role of the GPI factors in relocating TC clusters between the CTL and ZHT experiments.

While it is desirable to further quantify the relative percentage contribution of each GPI factor to the overall GPI change between the CTL and ZHT experiments as in, for example, Vu et al. (2021) or Murakami and Wang (2022), we should mention that their method of estimating percentage contributions does not perform well herein. This is because the GPI distributions between the two experiments are so different that any percentage estimation in the common incremental form no longer ensures that the sum of all GPI factor percentages is equal to the actual GPI changes.¹ As such, the absolute difference in each GPI factor is provided in Fig. 7 (contours) instead of the percentage contributions.

The above results highlight the due role of SST in global TC formation. At the storm-scale level, SST directly feeds surface enthalpy fluxes to TC development and theoretically results in a different TC intensity for different SST. At the

larger scale, the spatial distribution of SSTs can induce changes to the atmospheric circulations and structure, which affect the environmental conditions influencing where TCs can form such as VWS, moist convergence, static stability, or even the ITCZ migration. From this perspective, the results shown in Figs. 5–7 reveal two important points. First, they indicate that the direct SST effect does seem to play a large role in the clustering of TCG in the WP basin, because the WP cluster of TCG indeed disappears after removing the WP warm pool. On the other hand, the emergence of other TCG clusters in the Pacific Ocean related to changes in VWS, low-level moisture, and MPI factors also suggests that the indirect impact of SST on the atmospheric large-scale circulations and thermodynamics is of more importance than the direct SST effect in the clustering of global TC formation. The changes in GPI factors as seen in Fig. 7 must be ultimately linked to some modification in the large-scale tropical circulations under the zonally homogeneous SST condition to which we turn next.

c. Breakdown of the Walker circulation

As discussed in the previous section, the big shifts in VWS and RH in the western Pacific and the disappearance of MPI in the WP appear to be most critical for the formation of new TCG clusters in the ZHT experiment. To examine how these changes in the environmental conditions are related to the change in the large-scale circulation from a broader perspective, Fig. 8 shows the vertical cross section of the zonal overturning circulation cells for the typical NH summer (September) and SH summer (February) as obtained from the CTL and ZHT experiments.

For the CTL experiment, AM4 captures a typical characteristic of the Walker circulation in both hemisphere summers, with strong ascending motion in the WP basin and descending motion in the eastern Pacific between 10°S and 10°N (Figs. 8 and 9a,c). The strong ascending branch of the Walker circulation in the WP basin strengthens low-level moisture convergence, thus creating a favorable condition for TCG consistent with the high GPI and TCG density in the WP basin (cf. Figs. 6, 7). The seasonal migration of the CTL zonal circulation also accords with that of the ITCZ and accounts for the development of distinct TCG clusters as well as their seasonality in the Pacific Ocean as expected.

The zonal circulation in the ZHT experiment shows, however, a very different picture as compared with that in the CTL, with a breakdown of the tropical zonal overturning circulation into several smaller cells in both hemispheres. Specifically for the NH summer, the ZHT zonal circulation now displays several ascending and descending branches within the 10°–12°N band, with large subsidence developing right over the WP basin and a new ascending area shifting to the western central Pacific (Figs. 8b, 9b). This change in the zonally oriented circulation cell structure suppresses TC formation in the WP but promotes strong convergence or divergence at the lower or higher level, respectively, in the central Pacific, thus explaining also the overall smaller zonal VWS as well as

¹ This issue can be seen directly by writing out the incremental form of the GPI change as follows: $\delta\text{GPI}/\text{GPI} \approx \delta\text{RH}/\text{RH} + \delta\eta/\eta + \delta\text{MPI}/\text{MPI} + \delta\text{shear}/\text{shear}$, where the overbar denotes a mean state. Such an incremental analysis works best when the relative change between two experiments is much less than the mean, i.e., $\delta x \ll \bar{x}$, where $x \in \{\text{RH}, \eta, \text{MPI}, \text{shear}\}$. For our case $\delta x \sim \bar{x}$ for all four GPI factors and so $\delta\text{GPI}/\text{GPI}$ is no longer equal to $\delta\text{RH}/\text{RH} + \delta\eta/\eta + \delta\text{MPI}/\text{MPI} + \delta\text{shear}/\text{shear}$ as expected for the increment evaluation.

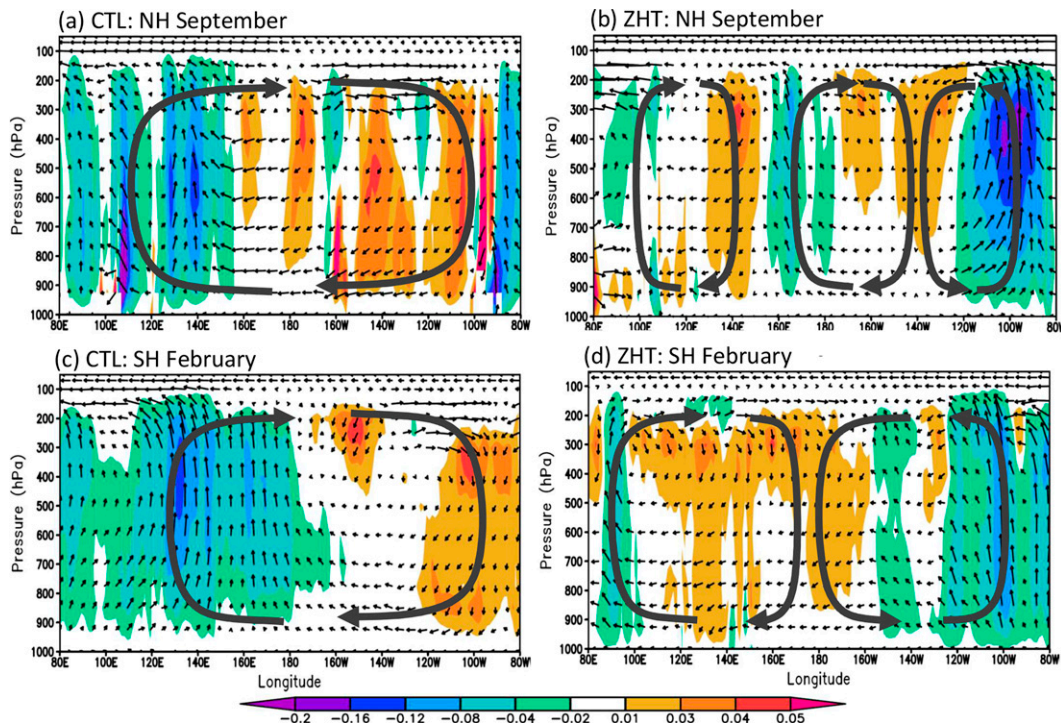


FIG. 8. Vertical cross sections of the vertical motion (shaded; Pa s^{-1}) and in-plane zonal flow (vectors; m s^{-1}) that are (top) averaged between 10° and 12°N in the NH and (bottom) averaged between 10° and 12°S in the SH for (a) CTL September, (b) ZHT September, (c) CTL February, and (d) ZHT February. The vertical motion component in each vector field is multiplied by a factor of 100 for displaying the zonally oriented circulation cells, and so the vector length does not reflect the actual magnitude of horizontal vs vertical velocity. Thick solid arrows show the dominant zonal overturning circulation cells in each experiment and hemisphere.

stronger low-level moisture environment near the central Pacific in the ZHT experiment.

A similar breakdown of the zonal overturning circulation with strong subsidence over the southwestern Pacific is also observed in the SH summer (Fig. 8d), although there seem to be fewer zonal cells in the SH (cf. Figs. 8b,d). These newly emerged zonal cells in the ZHT experiment are consistent with the distribution of TCG clusters shown in Fig. 4, thus revealing the important role of the large-scale circulation in modulating environmental conditions for TC development. Given that there are no TCG clusters in various aquaplanet global simulations, it is therefore expected that the breakdown of the zonal overturning circulation into smaller cells and the subsequent development of new TCG clusters must be rooted in the land surface arrangement or terrain forcings. Such an important role of land surface topography may also explain the difference in the ZHT zonal circulation cells between the SH and NH as shown in Fig. 8.

Also of interest from the ZHT experiment is that the clustering of global TC formation is not entirely associated with the ITCZ but strongly depends on the collocation of the large-scale rising motion and the warmest SST band. In the NH summer, the large-scale rising motion, the warmest SST band, and the ITCZ location in the ZHT experiment are all collocated during the TC peak season (June–October) as seen in Figs. 5b and 9b. During the SH peak TC season (January–

March), the ITCZ still resides in the Northern Hemisphere (Fig. 5d), yet the areas of rising motion and the warmest SST band are now around 8° – 12°S (Fig. 9d) where the clusters of TCG emerge (cf. Fig. 6h). Note that the ascending areas between 8° and 12°S during the SH summer are not a part of the ITCZ in the ZHT experiment, which is still located in the NH. In this regard, the ZHT experiment provides an ideal framework to separate the role of the ITCZ, the areas of large-scale rising motion, and the SST distribution in global TC formation that is otherwise difficult to examine from either the CTL or aquaplanet experiments.

From the global perspective, one can see now that the tropical circulations and SST play different roles in the TCG clustering; the former provides large-scale atmospheric conditions such as VWS, tropospheric static stability, moisture convergence, or vorticity-rich environment for TC formation, while the latter provides the surface air–sea fluxes needed for TC internal dynamics. In our idealized AM4 simulations, the response of the large-scale circulations to the land surface distribution is apparently more crucial for the longitudinal clustering of TCG, even in an environment with no zonal SST variations.

4. Sensitivity experiments

The last question that we address is the robustness of the results obtained from the ZHT experiment including the shift

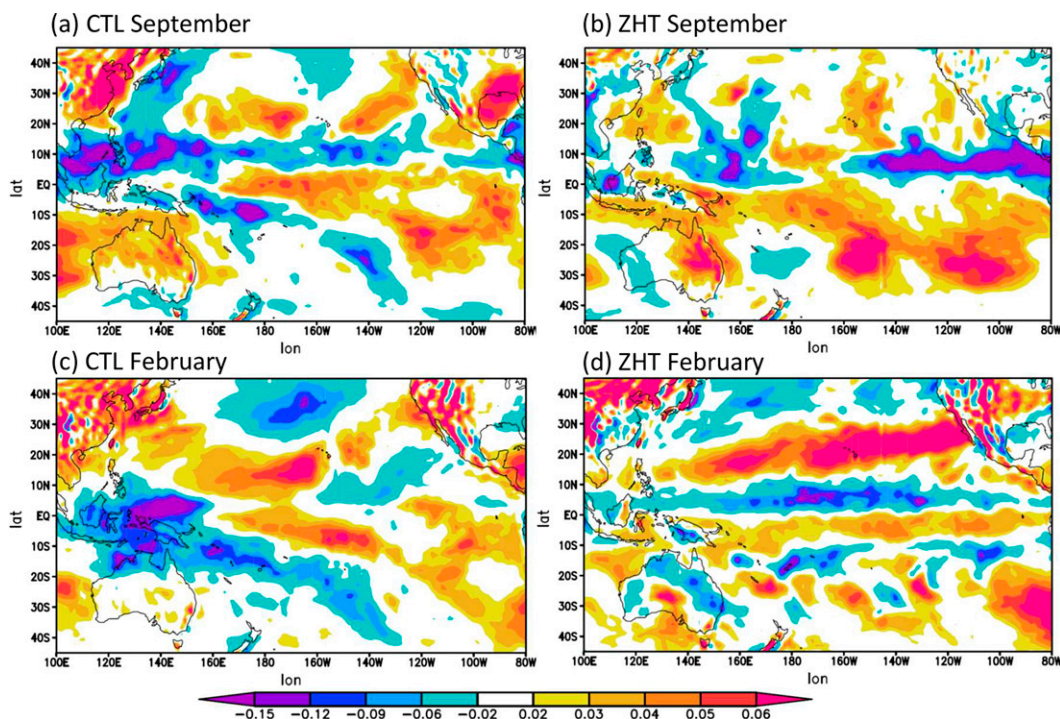


FIG. 9. Distribution of the vertical motion (shaded; Pa s^{-1}) at 500-hPa level over the Pacific Ocean as obtained from (a) CTL September, (b) ZHT September, (c) CTL February, and (d) ZHT February.

in the TCG clusters, the strong asymmetry of the TCG distribution between the two hemispheres, and the breakdown of the Walker circulations under the zonally homogeneous SST. In the following, we present two groups of sensitivity experiments outlined in section 2b to answer this question.

a. Global warming scenarios

In the first set of sensitivity experiments, we examine how different warming scenarios affect global TC formation and the large-scale tropical dynamics relative to the ZHT experiment. Figure 10a shows the time series of global TC frequency for three different warming scenarios listed in Table 1. One notices overall that the tropical warming scenario leads to a reduction in the global TC number relative to the ZHT experiment, similar to what was obtained in previous studies with more realistic SST distributions (Knutson et al. 1998, 2010, 2020; Murakami and Wang 2010; Hsieh et al. 2020; Walsh et al. 2020). The decrease in the global TC count is most apparent for the tropical warming scenario (-23%), but less significant in the polar and homogeneous warming experiments. This behavior of TC frequency suggests that a warming in the tropical region tends to be most effective at reducing global TC frequency when compared with other warming scenarios, and it may be relevant to previous findings using projected twenty-first-century climate change patterns (e.g., Knutson et al. 2020). The fact that warmer tropical SST climate leads to fewer TCs also suggests that it is the unfavorable changes in the large-scale tropical circulation that dominate global TC

formation response by opposing the direct effects of enhanced sea surface fluxes related to warmer local SSTs in the tropics.

In terms of the spatial distribution of TCG density in these warming scenarios, Figs. 10c and 10d show that the clustering of global TC formation is very persistent among all experiments, regardless of warming scenarios. Consistent with TC frequency, Figs. 10c and 10d show that the magnitude of TCG density is most affected in the tropical warming scenario, yet the pattern and distribution of TCG clusters is almost the same as those in the ZHT experiment. Similar analyses of the tropical zonal circulations in the Pacific Ocean confirm that the breakdown of the Walker circulation under these zonally homogeneous SST conditions is robust and accords with the overall clusters of TCG (Fig. 11). Comparison of Figs. 10 and 11 suggests also that the weakening of vertical motion within the ITCZ, which corresponds to weaker moisture convergence within the ITCZ and the increase in VWS within the main genesis region, account the most for the overall decrease of TCG in the warmer climate experiments, especially in the NH summer.

Recall that the role of large-scale vertical motion might be implicit in the GPI factors, because it plays a dual role in producing a moist environment and spinning up low-level vorticity via absolute angular momentum convergence. As discussed in Murakami and Wang (2010), Wang and Murakami (2020), and Hsieh et al. (2020), the role of vertical motion can be made more explicit in the TCG probability by including a factor that represents the 500-hPa ω field. For our sensitivity analyses in this study, however, we focus more on the relative

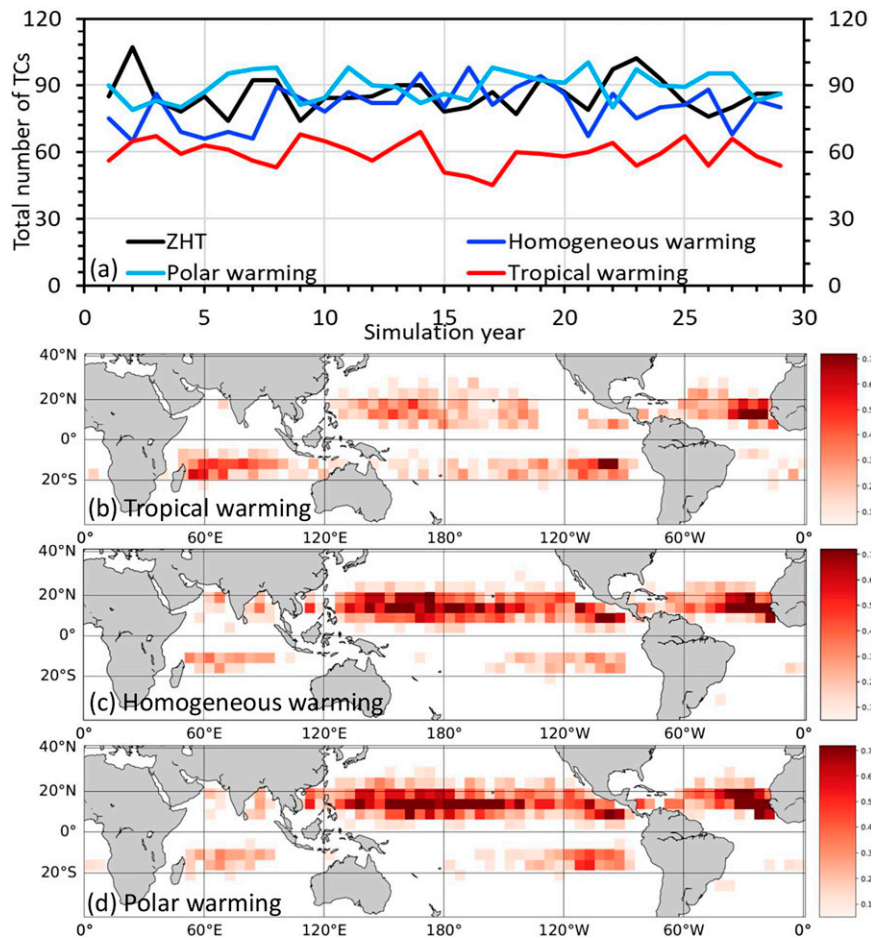


FIG. 10. (a) Time series of the global TC counts during the 30-yr simulations obtained from the ZHT experiment (black) and three different warming scenarios: tropical warming (red), homogeneous warming (blue), and polar warming (cyan). Also, as in Fig. 4b, but TC formation density for the (b) tropical warming, (c) homogeneous warming, and (d) polar warming experiments.

difference in the clustering of global TC formation among different warming scenarios. As such, no additional analyses of different GPI modifications are provided. Regardless of different GPI formulas, results from these sensitivity experiments not only confirm the robustness of our results but also provide some insight into the likely response of global TC formation under different warming scenarios.

b. Deep convection sensitivity

In the second set of sensitivity experiments, the AM4 deep convection parameterization was modified to see how the model physics affects the robustness of our analyses. Unlike the warming scenarios that focus on the sensitivity of TC formation to large-scale environments, these physics sensitivity experiments are to verify if model options could change any characteristics of the clustering of global TC climatology.

In this regard, Fig. 12 shows the global TC frequency and the distribution of TCG density for the simulations with deep convection entrainment rate reduced, which are applied for

both the CTL and ZHT experiments. One can see first from these sensitivity experiments that reducing the deep convection entrainment rate by 20%, that is, strengthening the parameterized deep convection, decreases the global TC count by about 10% in both CTL and ZHT settings. This is interesting, as deep convection is often considered to be the key ingredient for TC development; that is, stronger deep convection mechanisms would lead to higher overall TC intensity. However, the argument of higher intensity for stronger deep convection is generally applied only for a single storm development instead of global TC frequency. On the one hand, increasing parameterized deep convection in numerical models tends to reduce static instability and ambient moisture, which would negatively affect TCG as discussed in Zhao et al. (2012). On the other hand, it can also help organize convection at larger scale and could promote TC development. From this perspective, the AM4 model shows that the local conditions to support stronger intensity for a single storm do not generally translate to more TC formation overall.

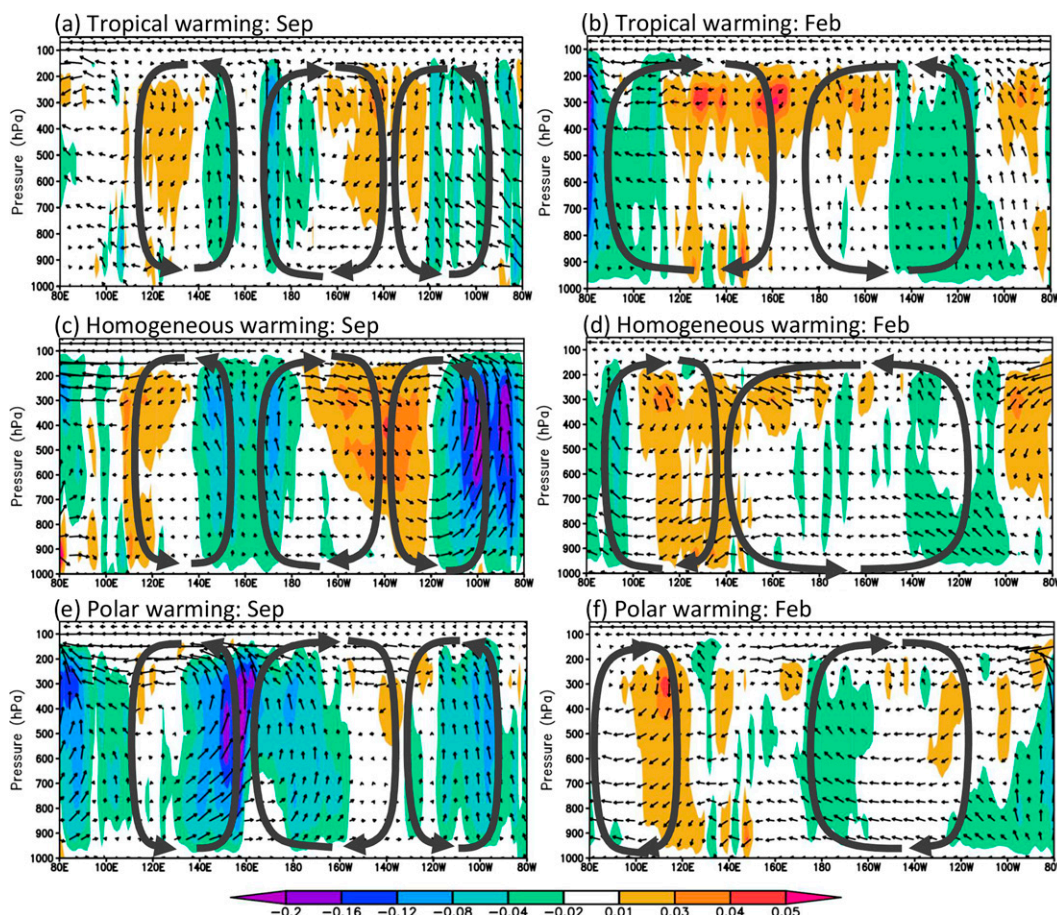


FIG. 11. As in Fig. 8, but for three warming scenario experiments (a),(b) tropical warming, (c),(d) homogeneous warming, and (e),(f) polar warming, beginning from the same zonally homogeneous SST distribution as in the ZHT experiment.

Although there are fewer TCs for both the control and ideal SST experiments, we observe again the same clustering of TCG in both CTL and ZHT settings even when the convection entrainment rate is modified (Fig. 12b). Consistent vertical cross sections of the zonal circulations as well as their breakdown are also observed in these sensitivity experiments, very similar to those obtained in the ZHT experiment (not shown). With these results, we are confident that our findings of the indirect SST effect on tropical large-scale circulation and related consequences on global TCG clustering in the presence of realistic land surface coverage and orography are robust features within AM4.

5. Conclusions

In this study, the direct and indirect roles of zonal variations of sea surface temperature on global TC formation were examined, using the global AM4 model. By designing a series of zonally homogeneous but latitudinally varying SST profiles in the presence of realistic land surface, a number of significant results were obtained from our series of 30-yr simulations. First, it was found that the modeled tropical atmosphere displays an

intrinsic variability in the global TC number with several dominant frequencies of 3, 6, and 9 years and an average fluctuation of 6–8 TCs even in the absence of all interannual SST variability and ocean coupling. Such intrinsic fluctuation is robust among all simulations regardless of idealized or realistic SST distribution and may have implications for the future TC projection studies. Specifically, this finding suggests that future projections or detection/attribution studies of global TC frequency may need to take into account intrinsic atmospheric variability as well as variability from ocean–atmosphere interactions such as ENSO (not modeled here) to assess internal variability “noise” levels before a greenhouse gas-induced trend in TC count can be confidently detected.

Second, our zonally homogeneous SST (ZHT) experiment simulated about 20% more TC formation as compared with the control simulation based on realistic SST distribution, with most of the increase in the Northern Hemisphere. Such an increase of global TC formation in the ZHT experiment is consistent with the more organized ITCZ that stays persistently in the Northern Hemisphere in the zonally homogeneous SST setting. In the ZHT experiment, the Northern Hemisphere has not only a larger warm ocean coverage but

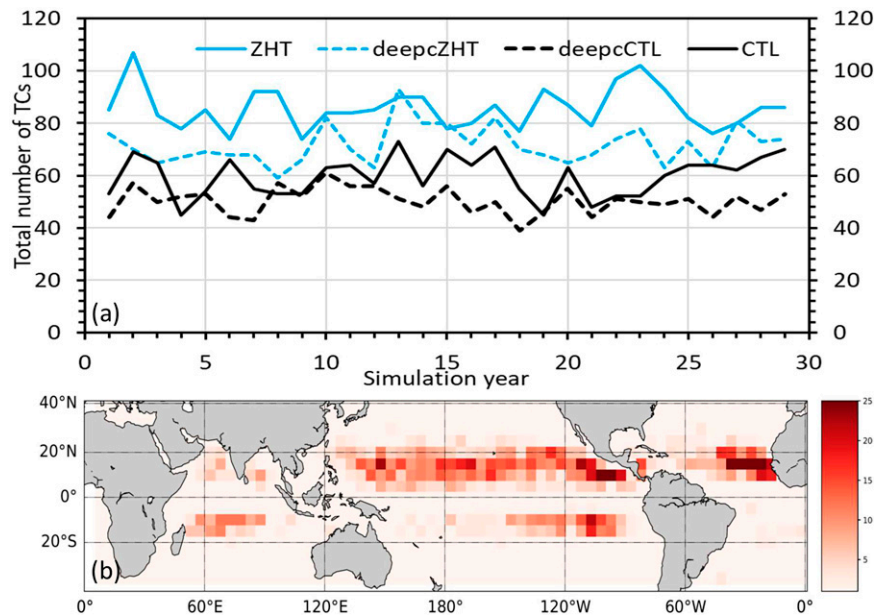


FIG. 12. (a) Time series of the global TC counts as obtained from the CTL and ZHT experiments with reduced deep convection detrainment (dashed black and blue lines, respectively) as compared with the time series of the global TC counts from the CTL and ZHT experiments (solid black and blue lines, respectively), respectively. (b) As in Fig. 4b, but for TC formation density for the reduced deep convection detrainment experiment using the same SST distribution as in the ZHT experiment.

also a higher chance of TCG associated with more organized ITCZ, thus producing more TCs in the Northern Hemisphere and globally overall.

Of more interest from the ZHT experiment is the emergence of new TC clusters across ocean basins, when compared with the CTL experiment, with the most drastic changes in the Pacific Ocean. Our examination of the genesis potential index captured similar new GPI clusters that are well aligned with the new TC clusters over all ocean basins under zonally homogeneous SST conditions. These major TC clusters are robust across sensitivity experiments with different warming scenarios or convective parameterization entrainment rate. In this regard, our result suggests that the direct surface enthalpy fluxes associated with zonal SST anomalies should not be considered the leading factor in the clustering of global TC formation.

To further examine the cause of TC clusters in the zonally homogeneous SST conditions, we analyzed the large-scale tropical circulation for different experiments. It was found that the clustering of global TCG in our ZHT experiment is a consequence of the breakdown of the Walker circulation in the presence of realistic terrain and land surface coverage. Specifically in the Pacific, the Walker circulation breaks into several smaller zonal cells in the tropical region under the zonally uniform SST distribution, with new ascending and descending centers across the Pacific Ocean. Due to different terrain forcing in different hemispheres, the organization of zonally oriented overturning circulations is different between the two hemispheres. However, the structure of the new zonal cells related to the zonal

circulation breakdown is consistent among all sensitivity experiments. Because of these new zonal cell structures, large-scale conditions such as VWS, low-tropospheric moisture, and potential intensity are all shifted, producing different TCG clusters even in the absence of all zonal SST anomalies. Thus, our results support that the response of tropical circulations to SST anomalies plays an indirect yet more important role in the clustering of TC formation at the global scale.

There are several caveats of our study to mention here. First, an apparent question of why the tropical atmosphere displays a specific interannual variability of 6–8 TCs at several dominant frequencies as shown in Fig. 2 is not currently understood. While we could not offer a good answer to this question with the AM4 simulations in this study, we note that the atmosphere is broadly a compressible system with a specific thermodynamic structure and elasticity. Thus, any large-scale forcing such as terrain features or seasonal variation of solar radiation likely causes the atmosphere to undergo natural internal variability. In this study, such an internal mode of global atmospheric oscillation is manifested in terms of global TC count, yet this global mode, to our knowledge, has not been understood or discussed in previous studies. To some extent, our result should be therefore considered as a report of what was obtained from the AM4 model herein, which needs further verification and more in-depth understanding.

Second, our study could not address why the breakdown of the Walker circulation in the ZHT experiment produces a zonal structure with a specific number of zonally oriented (east–west) circulation cells as inferred in Fig. 10. Understanding this

question would require more sensitivity analyses of large-scale circulation in the presence of different landmass coverage, terrain heights, or idealized SST profiles, which are, however, outside the scope of this work. Such sensitivity experiments are needed not only for understanding the specific behaviors of global TC clusters, but also to help understand the migration of ITCZ and monsoon activities as discussed in [Geen et al. \(2020\)](#). We emphasize, however, that the lack of TC clusters as well as the lack of such multicell zonal structures in global aquaplanet simulations suggests that the root of the breakdown of the Walker circulation into these multiple cells must be related to the land surface/terrain forcings in the absence of zonal SST variation. These terrain forcings have been known to trigger some internal stationary modes of tropical dynamics from previous studies ([Wang et al. 2019](#)) that can be examined using idealized framework. More investigation into this type of external forcings that set up large-scale “hotspots” for the clustering of global TC formation is currently underway and will be presented in our upcoming study.

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Data availability statement. All model postprocessed time series and related data used to produce the results presented in this study can be accessed online (<https://doi.org/10.13140/RG.2.2.16128.92160>).

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